

Where retrieval-grounded QA breaks — and what should happen next

Citation-grounded question answering over EU aviation regulations

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Regulation (EU) No 376/2014 — occurrence reporting

The original 63/100 mixed system failure with data defects

82/100

Designated legacy chunk retrieved

63/100

Loose proxy threshold passed

43/100

Exact citation + content

But the same baseline also contained:

- flat chunks up to 30,275 characters
- a fixed 1,500-character context truncation
- parser and document-identity defects
- unchecked synthetic gold answers
- article-level partial citations counted as 0.5

**The original number is a baseline,
not a clean RAG ceiling.**

The repaired pipeline isolates data defects before attributing anything to the model.

The useful question is not “which system wins?”

BASELINE QUESTION

Which performs best: few-shot, RAG, or LoRA?

REFRAMED QUESTION

Where does retrieval-grounded QA fail, and which observable condition justifies escalating?

Target contribution

Use the simplest intervention that recovers the failure. Do not apply an agent to every question.

1

Auditable

A compliance reviewer can inspect why the system escalated.

2

Economical

Cost is spent only on the questions that need it.

3

Falsifiable

Escalation precision and recall make the claim testable.

Every candidate answer is now linked to exact legal evidence

109

Legal parent units

1,535

Retrievable child units

99

Retained candidate questions

42 / 57

Parent-grouped dev / test

1

Hierarchy preserved

Articles, annexes, recitals and guidance kept distinct; 5 instruments never collide.

2

Grouped splitting

Partitioned by legal parent, not by question ID. Zero parent leakage.

3

Evidence-linked gold

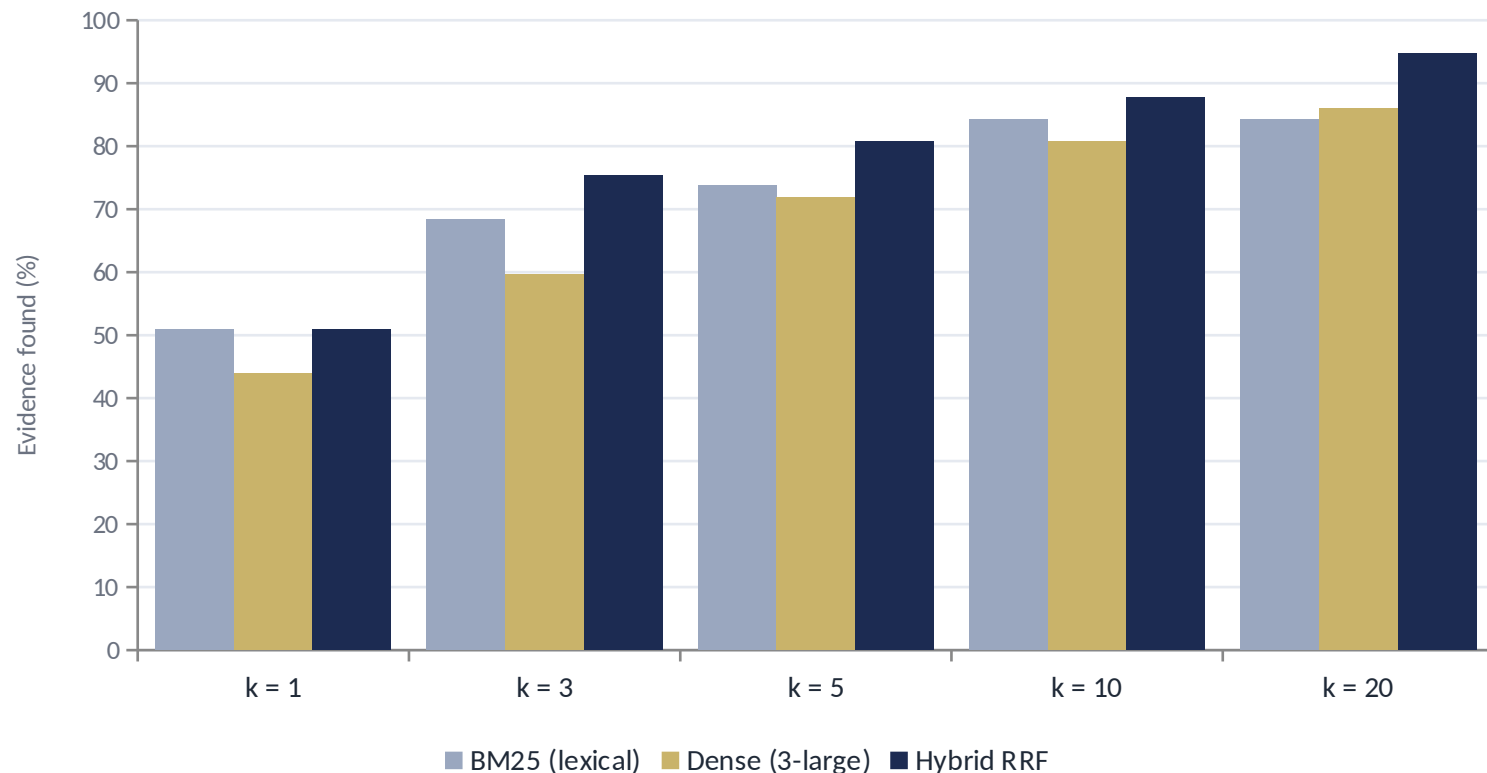
21 records corrected, 1 invalid record excluded, every record tied to child IDs.

4

Deterministic rebuild

SHA-256 lineage; a second independent build produced identical bytes.

Neither channel alone reaches the evidence



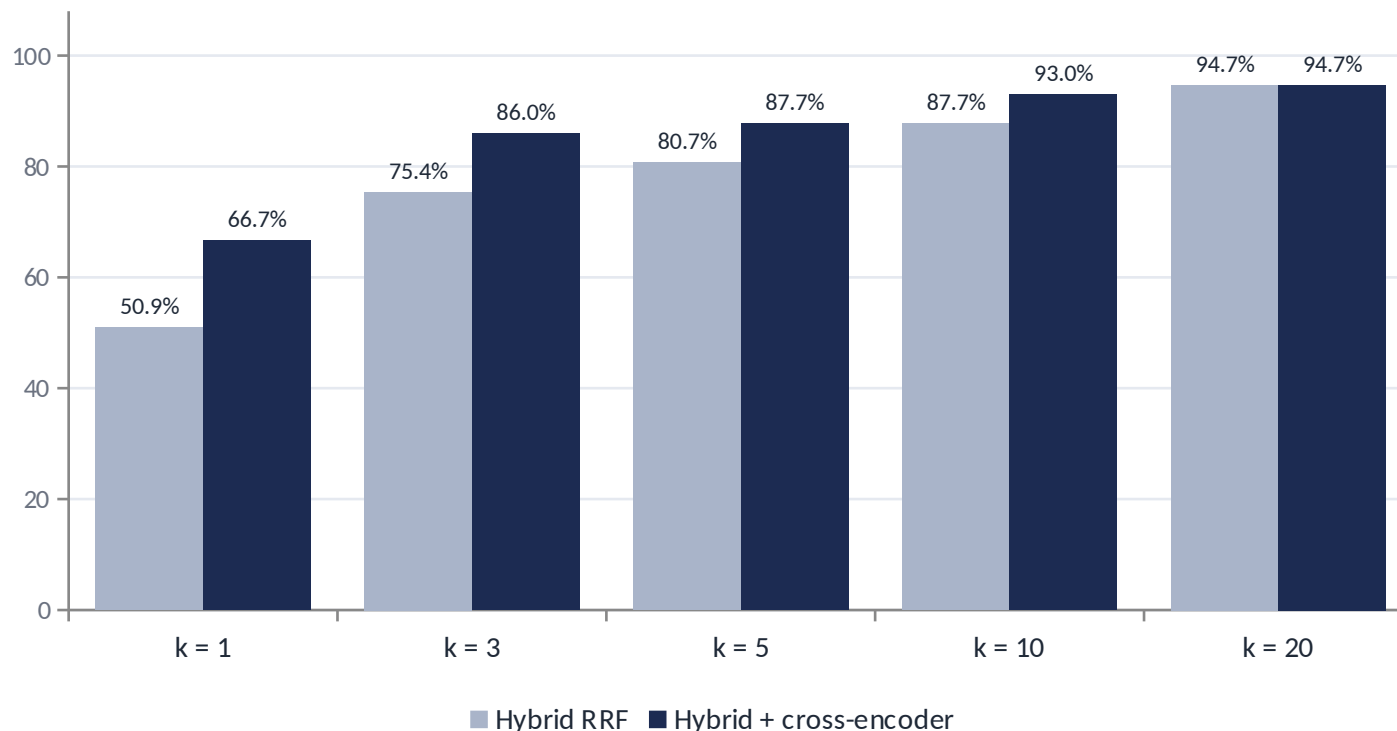
94.7%

Hybrid finds selected evidence for 54 of 57 test questions at k = 20

Why both channels

- BM25 wins on article numbers, acronyms and repeated legal wording
- Dense wins on paraphrase with no shared vocabulary
- Fusion is equal-weight, constant fixed at 60, never tuned on test

Reranking turns candidates into a usable context



50.9 → 66.7%

Top-1 accuracy after reranking

87.7%

Evidence inside a 5-source context

A reranker repairs ordering. It cannot recover evidence that was never in the pool.

Three remaining misses are three different problems

Reporting them as one “system ceiling” would be a measurement error, not a finding.

QA_0063

Clarify

“What exceptions?” is too broad — several real exception provisions compete for the answer.

The question is ambiguous.

QA_0080

Repair the gold label

Equivalent annex wording appears outside the single selected child.

The benchmark is incomplete.

QA_0176

Improve retrieval

A broad paraphrase does not uniquely identify the intended guidance evidence.

A genuine retrieval failure.

Independent review must separate false metric failures from real system failures before any ceiling is quoted.

The benchmark cannot certify itself

The problem

The same AI-assisted process that proposed the corrections cannot confirm them.

- 33 records: 22 audit exceptions, 3 residuals, 10 stratified controls
- Required IDs pinned to input hashes — the review cannot drift to another data version
- The freeze command refuses to run on an incomplete review

The protocol, fixed before collection

1

Multi-reviewer panel

Practising aviation professionals, not the author alone.

2

Pre-registered rule

Minimum two raters; strict majority on the decision and all four checks.

3

Corrections stay with me

The panel flags a problem; a multiple-choice form never writes gold data.

4

Agreement is reported

Cohen's or Fleiss' κ , plus every contested record listed individually.

Disagreement between qualified reviewers is a result, not noise — it identifies the ambiguous questions.

First returns: the protocol works, and it already found things

2

Reviewers, both 10+ years in aviation

27 / 33

Records rated at least once

7

Records rated twice — the overlap

K = 0.32

Fair agreement, n = 7

QA_0063 confirmed

Both reviewers independently voted to exclude it.

I argued from internal evidence that this question was ambiguous. Two aviation professionals reached the same conclusion without being told the hypothesis.

An unplanned finding

Six records drew the same reaction: the citation label itself is unreadable to a practitioner.

“GM 3.1 = ?” · “GM 4.2 = ?”

“what do you mean by section 2.2?”

“what do you mean by Annex II, item 3.14?”

This is a defect in my citation formatting, not in individual records — and no internal metric would have surfaced it.

Escalate only when the current evidence justifies it

0–1	Hierarchy-aware retrieval	Default after corpus repair
2	Hybrid candidates, top-20	Low fused score or channel disagreement
3	Cross-encoder reranking to ~5	Small margin between top candidates
4	Evidence-constrained generation	Source IDs, verbatim quotes, no model-written citations
5	Clarify or expand	Verification failure, or several plausible legal readings
6	Abstain → human review	Conflicting sources, no supporting evidence, unresolved ambiguity

Every trigger must be computable at runtime. Gold-label agreement is unavailable in deployment and cannot appear in a trigger.

Where the project stands today

Complete and reproducible

- Hierarchy-aware corpus: 109 parents, 1,535 children, zero validation issues
- Evidence-linked 99-question benchmark, parent-grouped, zero leakage
- E1–E4 complete: 3-small, 3-large, BM25, RRF, local cross-encoder
- Both context variants built for all 99 questions
- Answer schema, prompt contract, quote verification, citation resolution
- Chapters 1–4 drafted; bibliography at 64 verified references
- Validity protocol registered, form live, first returns aggregated

Blocked, and why

- Review collection: 27/33 rated once, only 7 rated twice
- End-to-end generation (E5) — blocked until the benchmark is frozen
- Verifier, abstention and switching-policy results
- Prototype interface

Freeze on 6 August on whatever coverage exists. Rule fixed in advance.

Revised plan — full draft by 12 August

The draft target moved forward by nine days. Experiments now freeze around 5–6 August.



Consequence: this argues for dropping the third embedding candidate rather than adding it, and for leaving multi-step retrieval unstarted unless a reviewed residual demands it.

Mémoire deadline: end of August.

Four answers unblock the final two weeks

1

Review coverage

Two reviewers so far, only 7 records rated twice. Is that overlap enough, or should I re-issue the form asking specifically for the 26 single-rated records?

2

Scope, and one clarification

By “the three models”, did you mean the three embedding candidates, or few-shot / RAG / LoRA? If the latter, I can re-score all three on the repaired benchmark with no new model calls.

3

Architecture

Approve hybrid top-20 plus local reranking as the retrieval architecture.

4

Draft review date

A slot shortly after 12 August, so feedback lands before the end-of-month deadline.

Next output: frozen gold benchmark → evidence-constrained answers → verifier and abstention results → switching-policy comparison.

Multi-step retrieval: the gate, and what would open it

The decision

Agentic retrieval is admitted only for a residual failure shown to require multi-step evidence.

Registered as D005 on 17 July, following the earlier framing of agents as a future direction.

What would open the gate — 2 to 3 days

- Build a cross-reference evaluation subset whose questions span two or more legal parents
- Show that reranking, expansion and clarification leave a residual on it
- Run a constrained multi-step policy on that residual; measure recovery, latency and cost
- Decision point: 6 August, once generation and verification are done

The measurement that settles it

99 / 99

Records whose gold evidence sits in a single legal parent

0

Records requiring two or more parents

Only two questions are typed as cross-reference, and both resolve within one parent. No question in the current benchmark requires multi-step retrieval — so an agent's benefit cannot be exhibited on this evaluation set, whatever its merits.

Either outcome is a result: agents shown unnecessary, or shown to recover a measured residual.